

D3 - SYSTEM MODELS PACKAGE

AgriUAV Platform: Precision Drone Dispatching Platform in the Mekong Delta

Subject	ITA301 - Information System Analysis & Design
Deliverable	D3 - System Models Package (30%)
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1. Diagram Package Index (Placeholders)

The table below indexes every UML/DFD artifact required for D3. Diagrams are intentionally left blank in this document - insert the rendered image (PlantUML render or draw.io export) at each placeholder from the listed source file. Linked UC(s)/FR(s) are restated so each inserted diagram can be checked against the rest of this package before final submission.

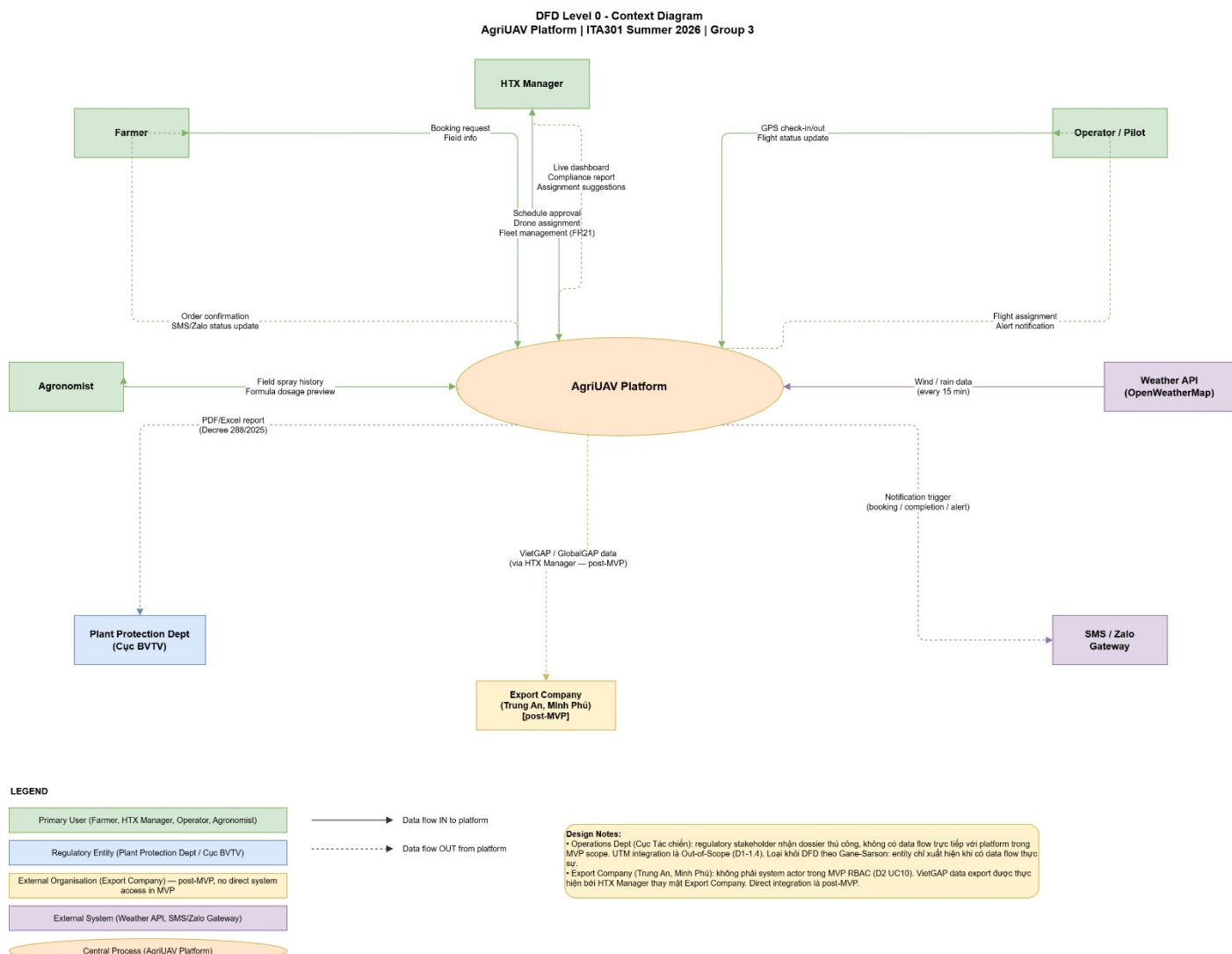
#	Diagram	Source File	Status
1.1	DFD Level 0 - Context Diagram	DFD_Agruav_level0_v1.5.drawio	Ready to insert
1.2	DFD Level 1 - 4 Sub processes	DFD_Agruav_level1_v1.5.drawio	Ready to insert
1.3	Use Case Diagram	D3.mdj	Ready to insert
1.4	Class Diagram	D3.mdj	Ready to insert
1.5	Activity Diagram 1 Booking Flow	D3.mdj	Ready to insert
1.6	Activity Diagram 2 Monitoring & Alert	D3.mdj	Ready to insert
1.7	Sequence Diagram UC01 Book a Spraying Session	D3.mdj	Ready to insert
1.8	Sequence Diagram UC03 On- site GPS Check-in	D3.mdj	NEEDS REVISION before insertion
1.9	Sequence Diagram UC04 Near Real- time Monitoring	D3.mdj	Ready to insert

1.1 DFD Level 0 - Context Diagram

Linked UC(s): - (context- level)

Linked FR(s): Scope- level (D1 |1.2 / D2 |1.2)

Shows AgriUAV Platform as a single process bounded by 8 external entities (Farmer, HTX Manager, Operator/Pilot, Agronomist, Plant Protection Dept, Export Company [post- MVP], Weather API, SMS/Zalo Gateway).



Balances against Level 0 with P1 Booking & Schedule Management, P2 Fleet Dispatch & Assignment, P3 Real-time Monitoring & Alert, P4 Compliance Logging & Report Export, and 4 data stores (DS1–DS4). Verify all 8 external entities and data flows n1–n5 are present before insertion.

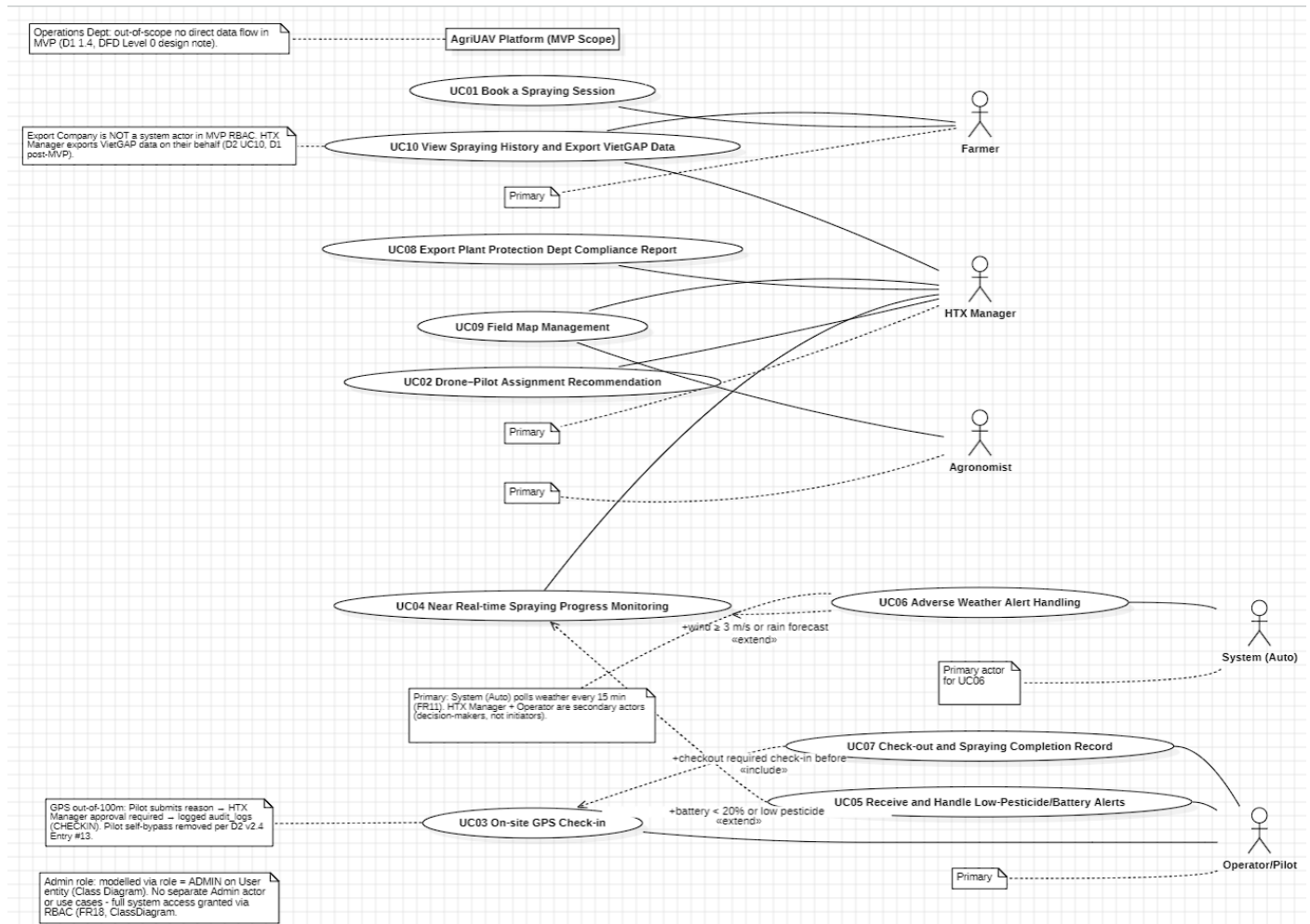


1.3 Use Case Diagram

Linked UC(s): UC01–UC10

Linked FR(s): FR01–FR21 (via UC, see [5])

Shows Admin generalizing HTX Manager (no separate Admin use cases), System (Auto) as primary actor for UC06, and «extend» (UC05→UC04, UC06→UC04) / «include» (UC07→UC03) relationships.

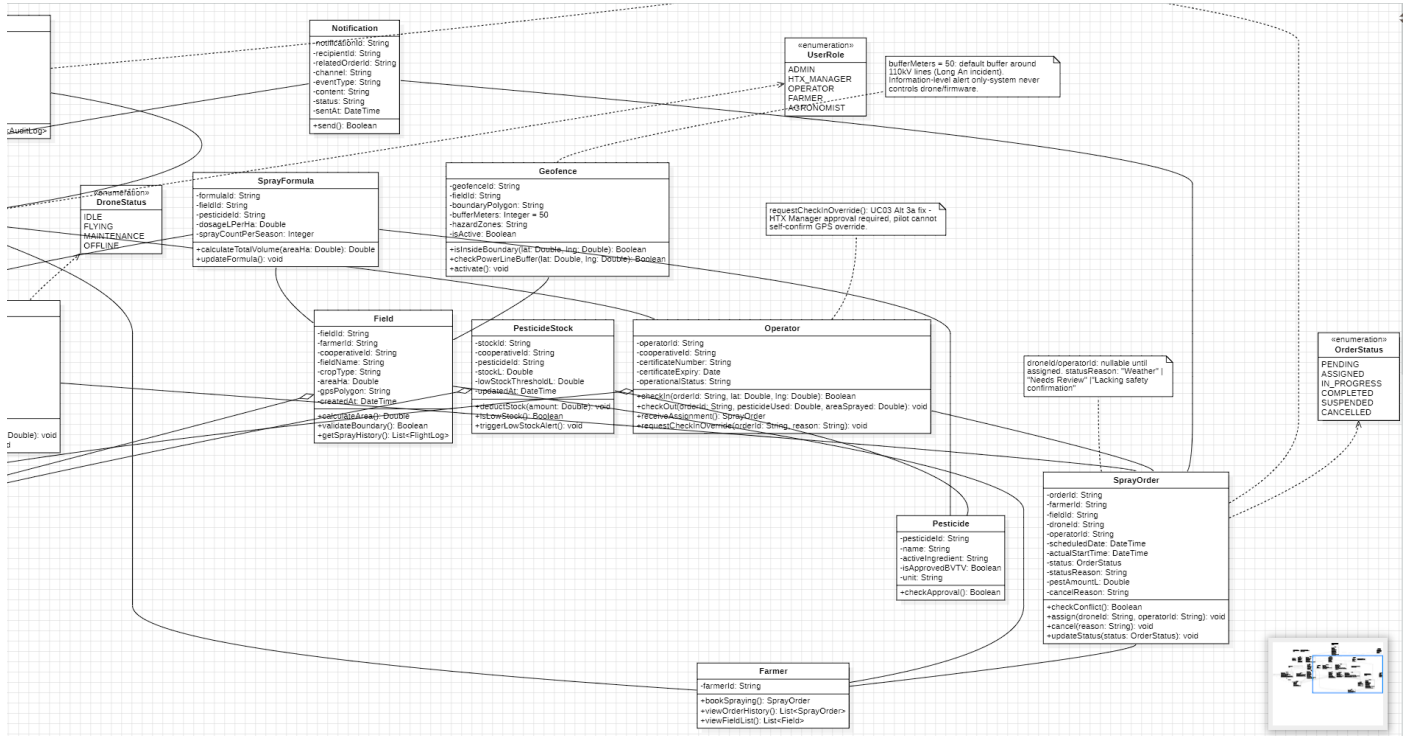
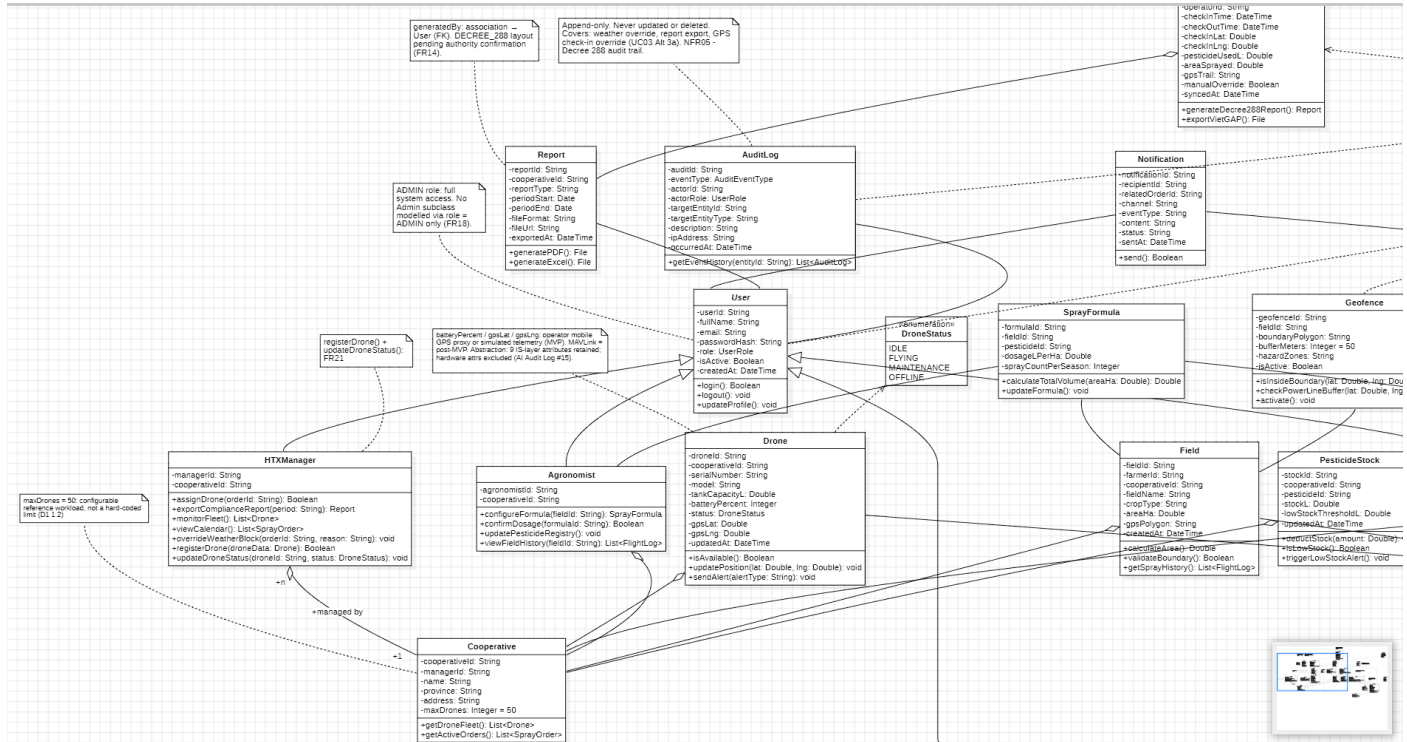


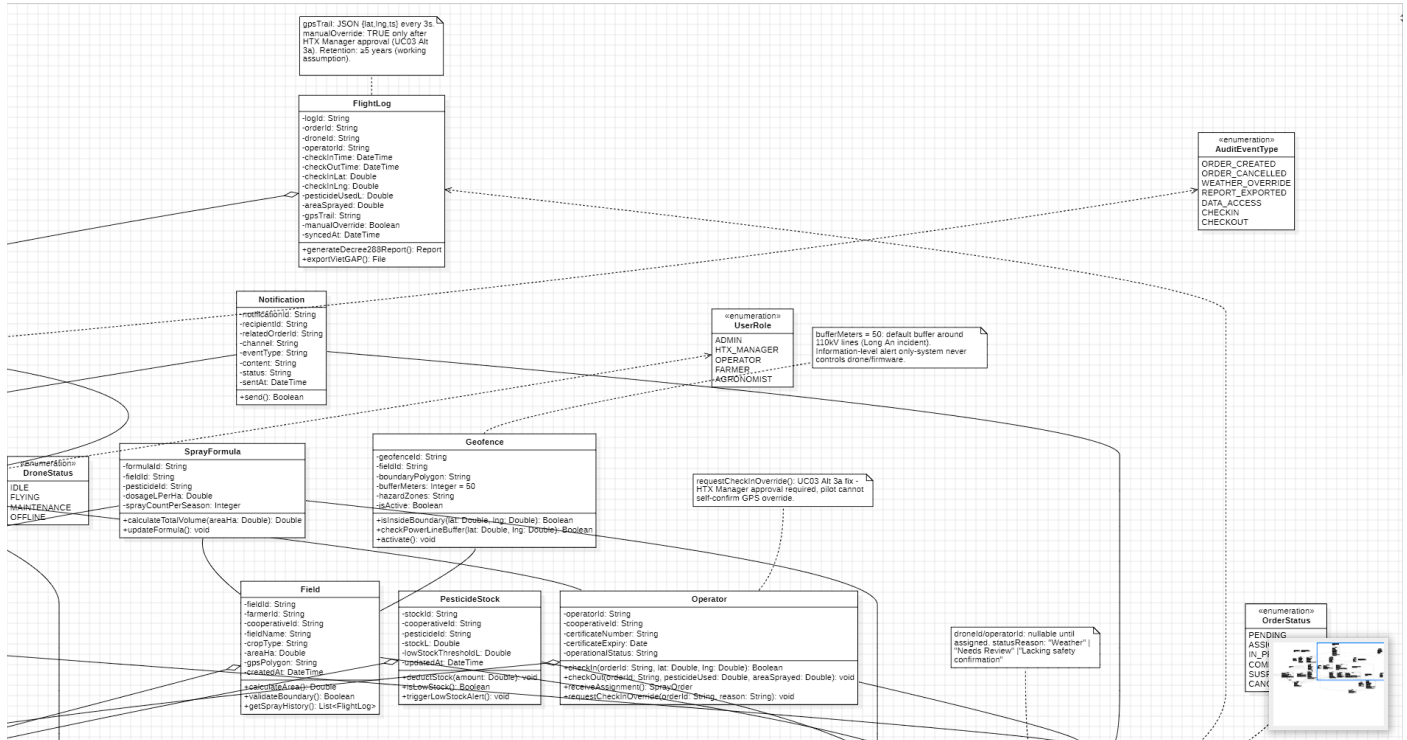
1.4 Class Diagram

Linked UC(s): All UCs

Linked FR(s): FR01–FR18, FR21

17 classes + 4 enums. Confirms the static structure backing | 2–4 of this document: User hierarchy (Farmer/Operator/HTXManager/Agronomist), Cooperative aggregation, SprayOrder lifecycle, FlightLog/AuditLog compliance pair, and the FR21 drone- registry methods on HTXManager.



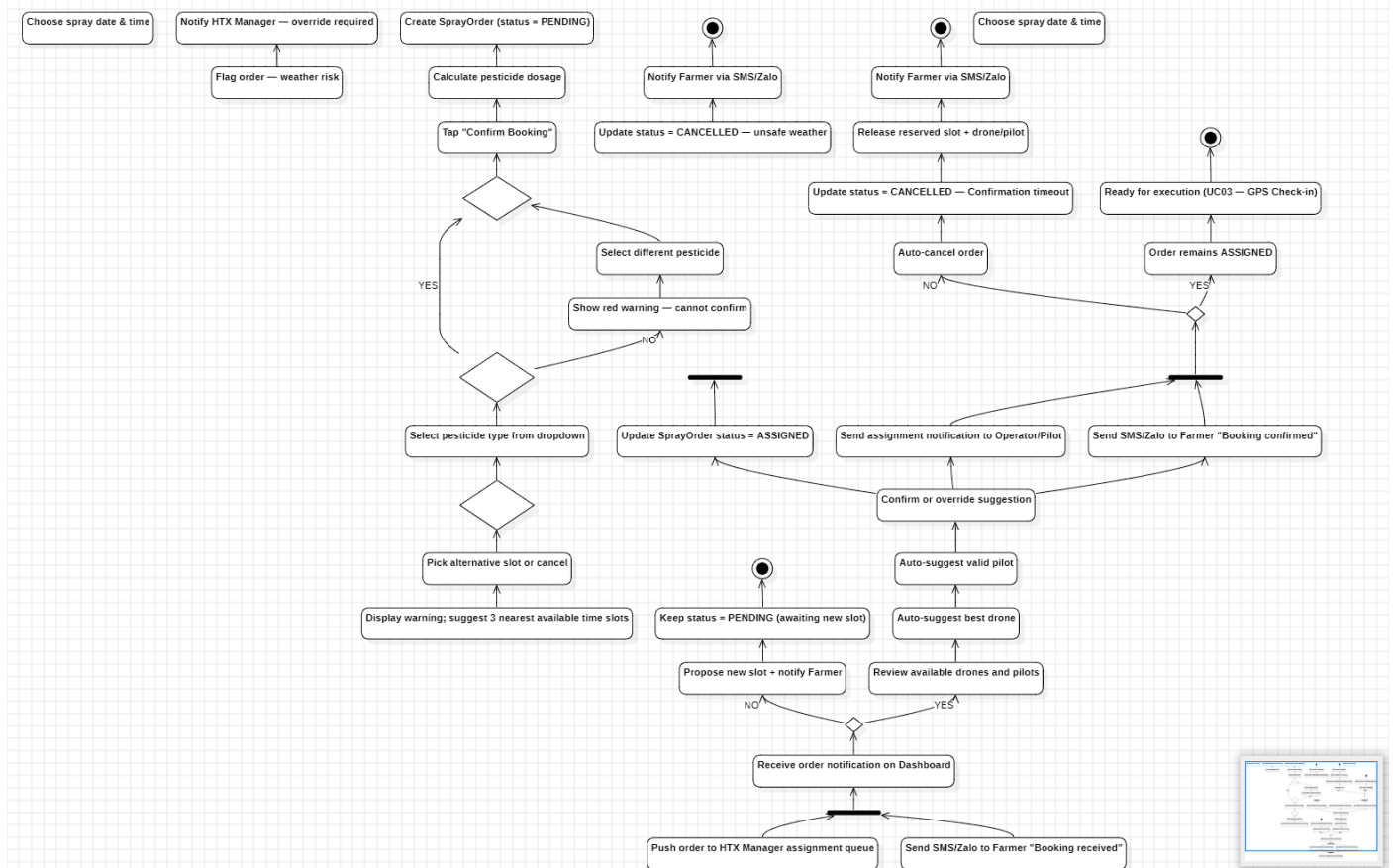


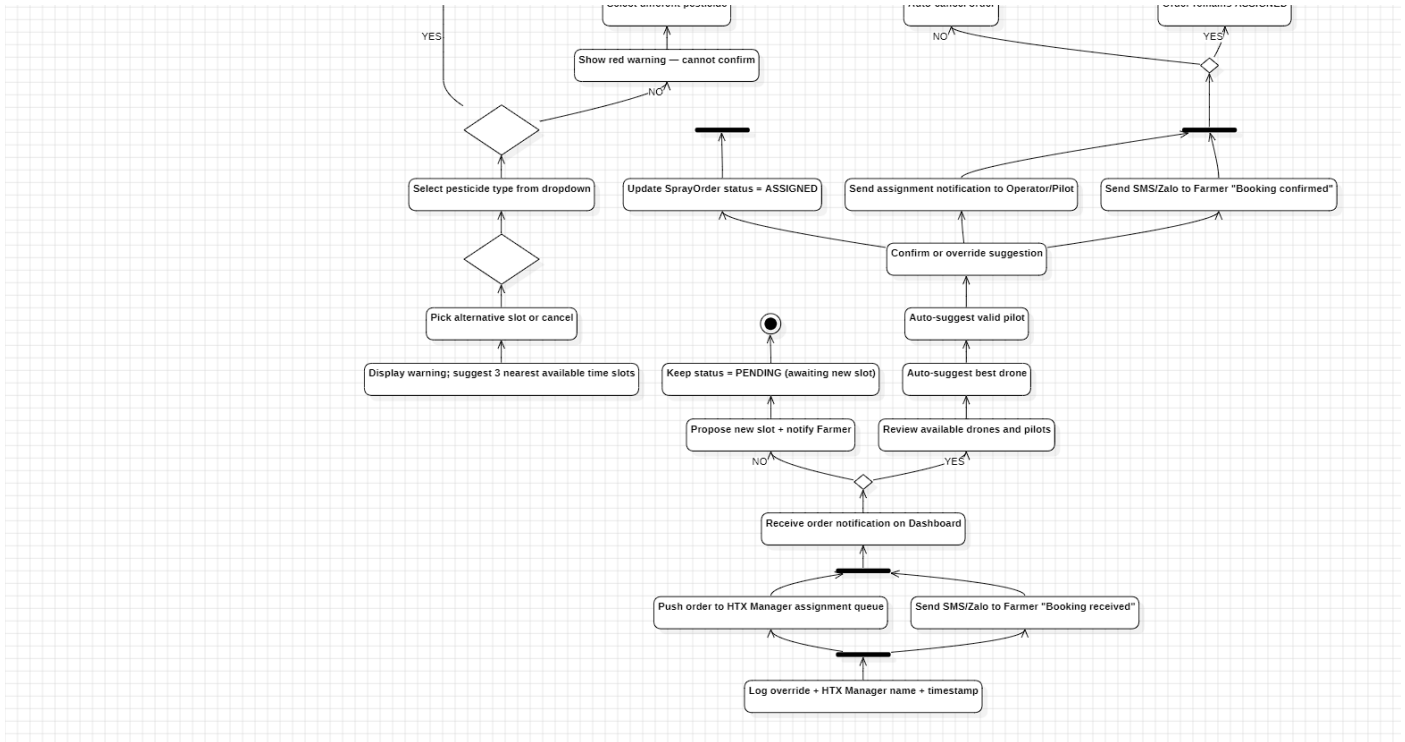
1.5 Activity Diagram 1 - Spray Order Booking Flow

Linked UC(s): UC01, UC02

Linked FR(s): FR01,02,03,04,06,07,08,13,16,17,18

3 swim lanes (Farmer / System / HTX Manager). Includes the UC01 weather soft- constraint gate with HTX Manager async override, and the corrected “no- assign” path that keeps the order PENDING rather than CANCELLED (UC02 Alt 3a).



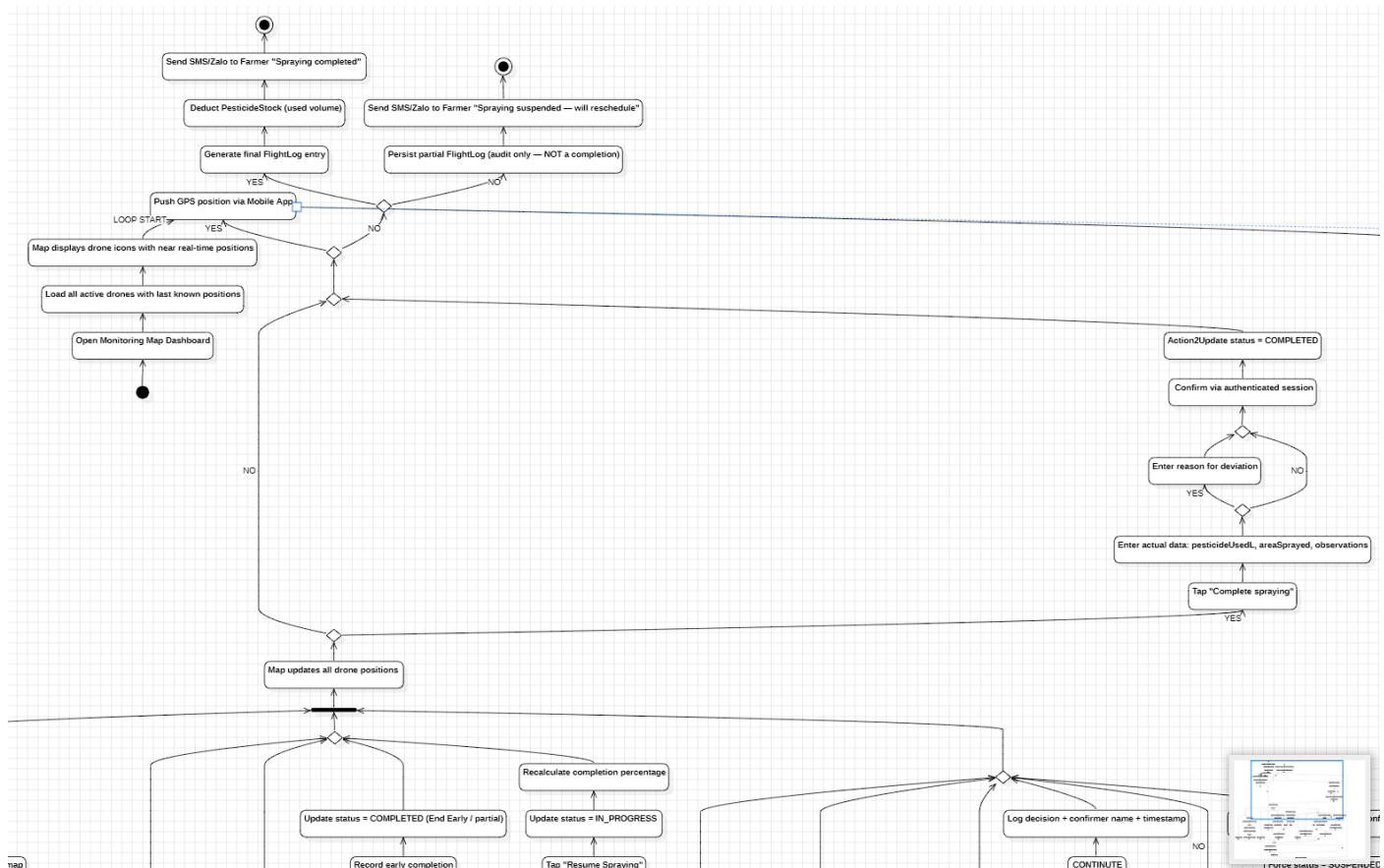


1.6 Activity Diagram 2 - Monitoring & Alert Handling Flow

Linked UC(s): UC04, UC05, UC06, UC07

Linked FR(s): FR09,10,11,12,13,16,17

3- way parallel fork (geofence / battery- pesticide / weather) plus the UC04 connection- lost check and UC07 check- out exit branch. When redrawing, replace any "GEOFENCE VIOLATION" label with "geofence alert" / "boundary crossing" wording

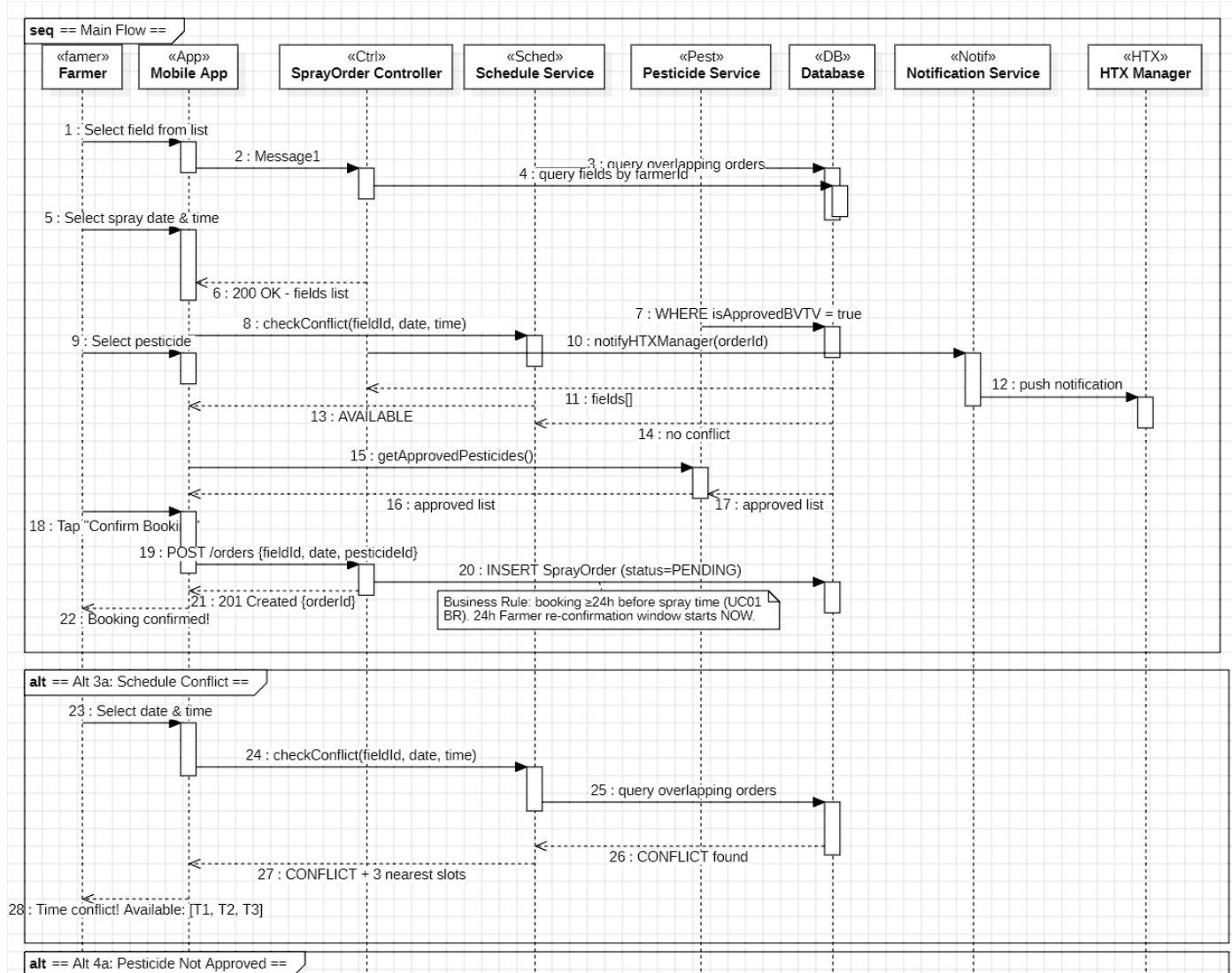


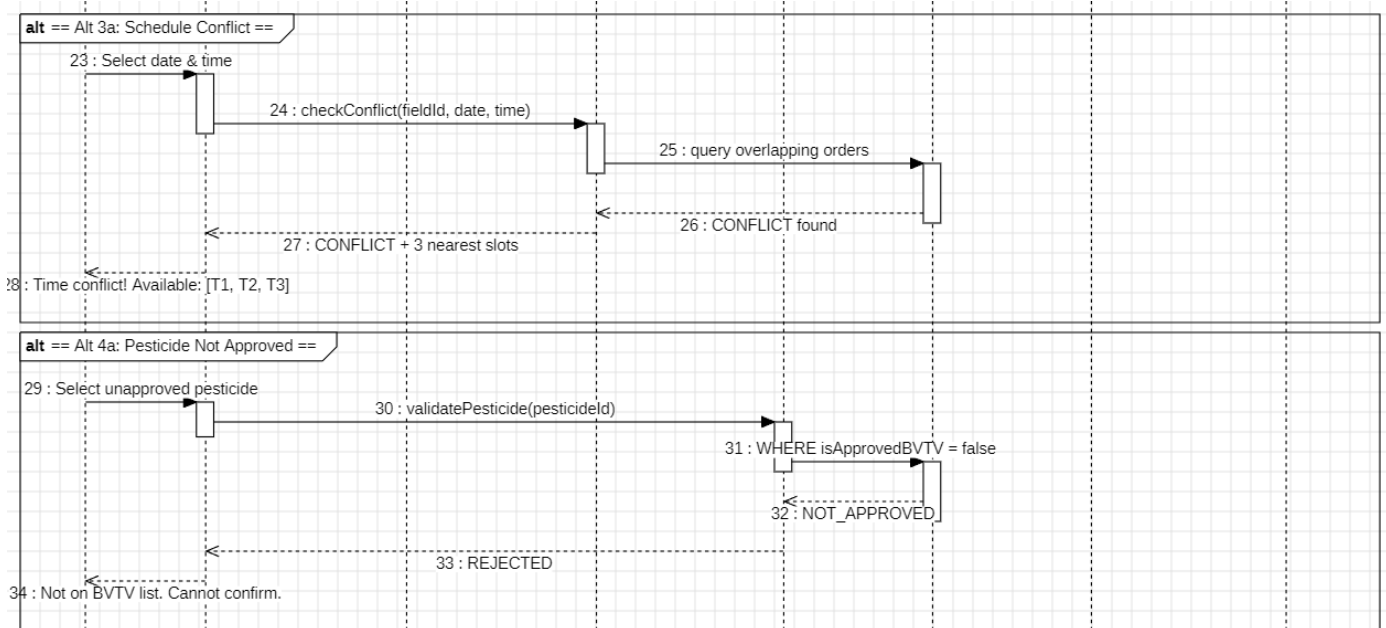
1.7 Sequence Diagram - UC01 Book a Spraying Session

Linked UC(s): UC01

Linked FR(s): FR03,04,16,17

Happy path + Alt 3a (schedule conflict) + Alt 4a (pesticide not BVTv- approved). No structural change needed for v1.5; carry forward from D3 v1.4 | 2.2.1.



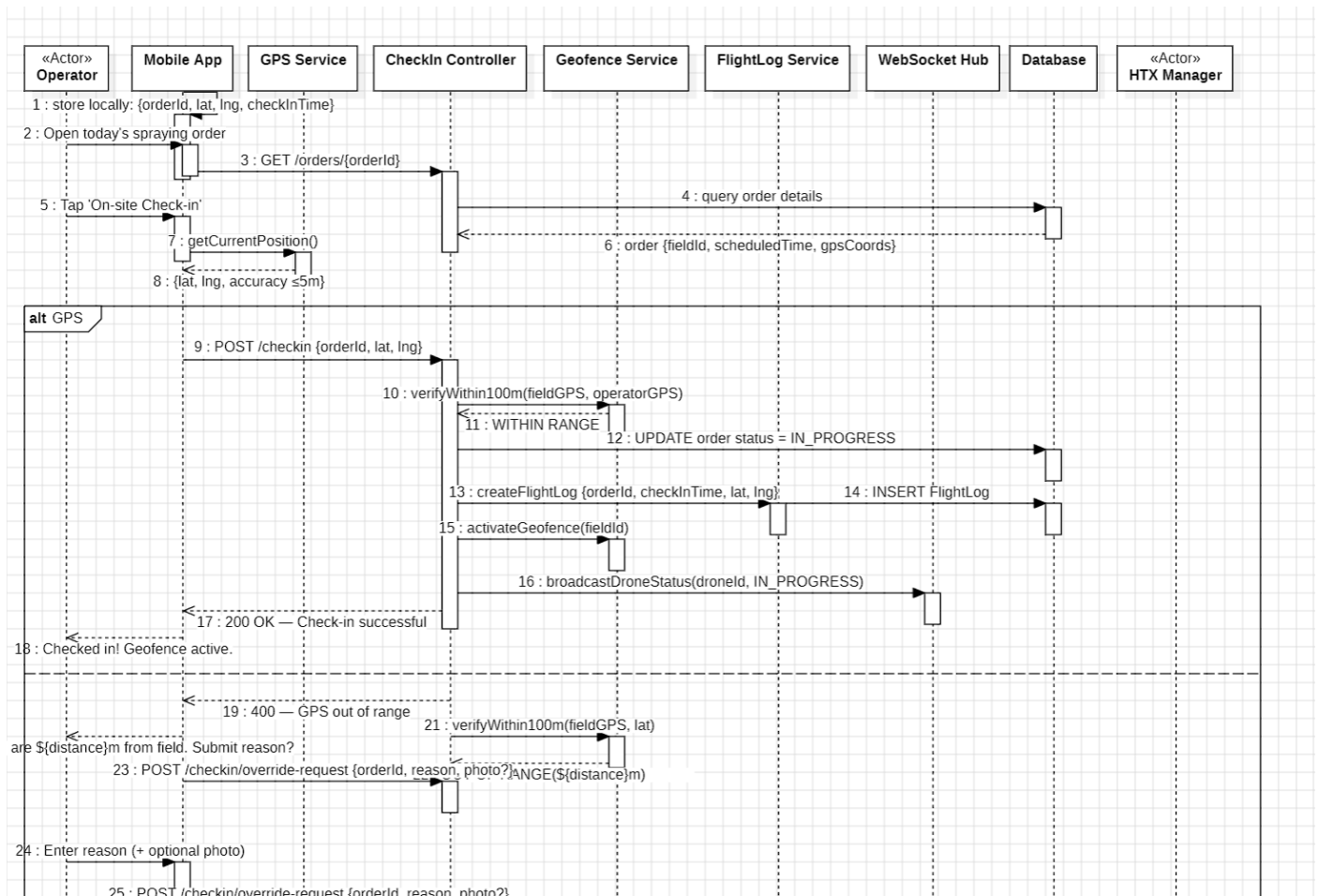


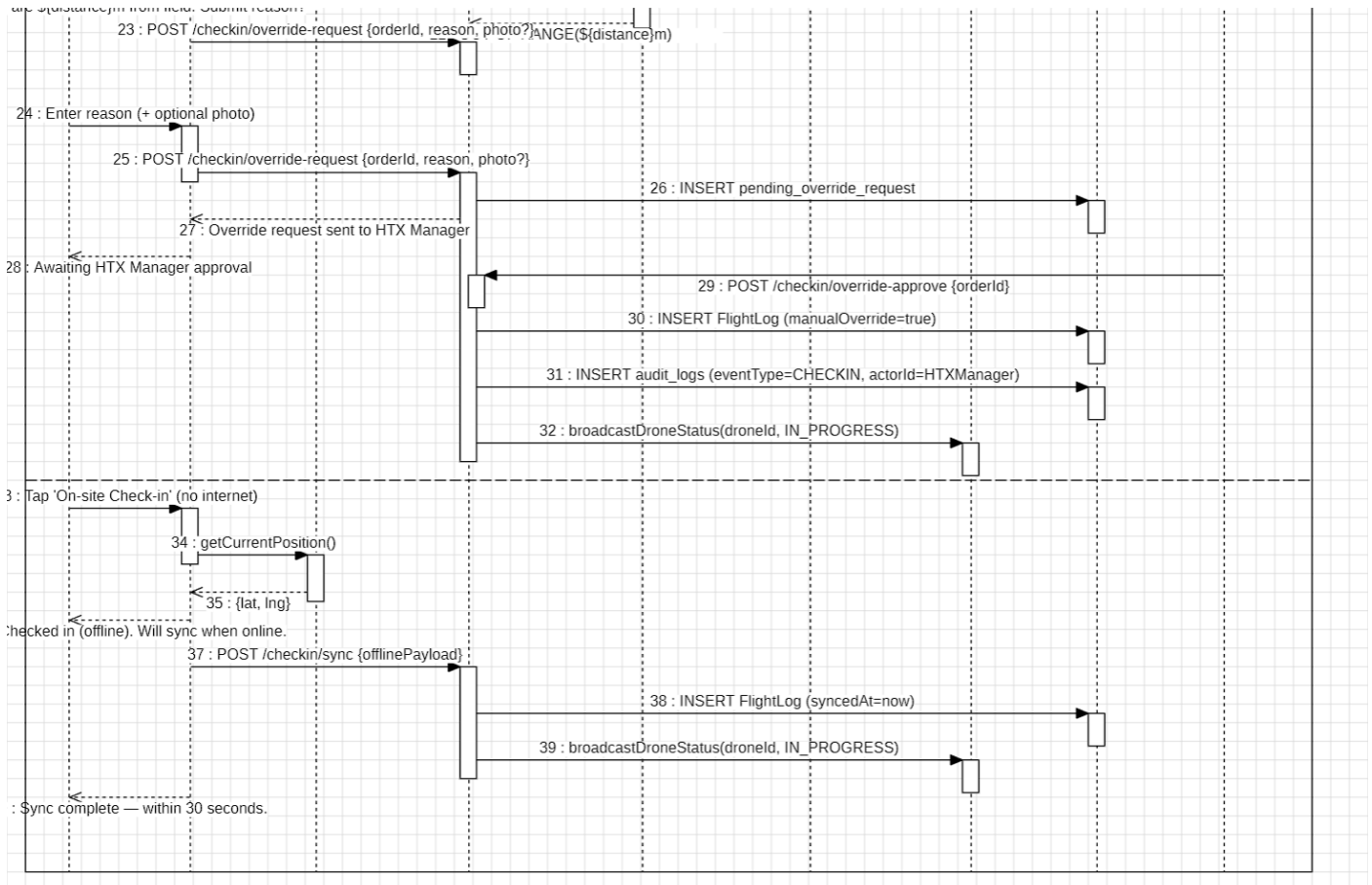
1.8 Sequence Diagram - UC03 On- site GPS Check- in

Linked UC(s): UC03

Linked FR(s): FR09, FR10, FR12, FR13

Happy path + Alt 3a (GPS outside 100m) + Alt 4a (offline / no 4G, IndexedDB cache, ≤30s sync per NFR06).



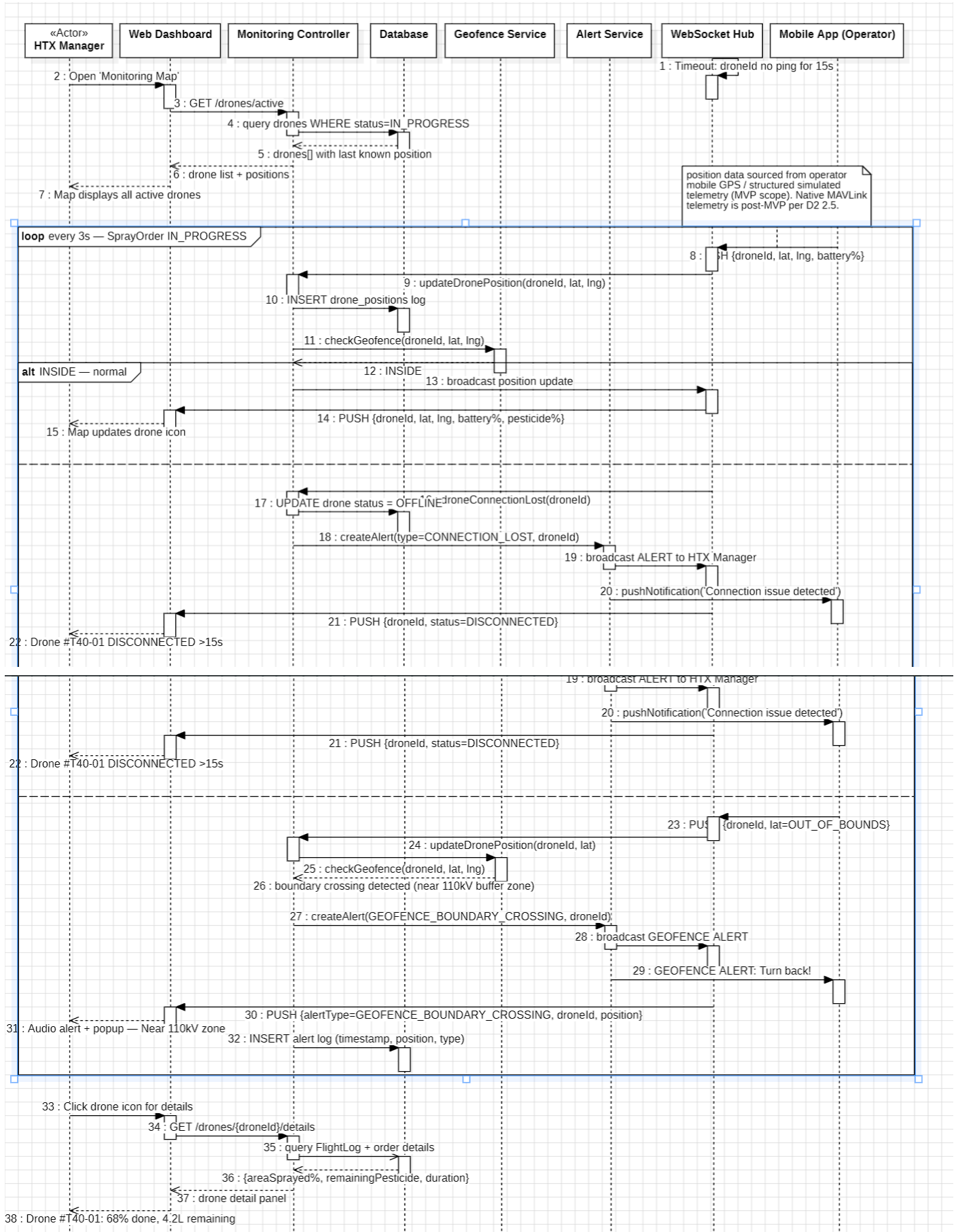


1.9 Sequence Diagram - UC04 Near Real- time Spraying Progress Monitoring

Linked UC(s): UC04

Linked FR(s): FR10,11,12,13

WebSocket loop + Alt 2a (connection lost >15s) + Alt 5a (geofence boundary crossing). When redrawing, use “information- level alert” / “boundary crossing” wording rather than “VIOLATION”, and “near real- time” rather than “real- time”, consistent with D2 v2.4 FR10/FR12.



2. Entity- Relationship Diagram - Logical Model

The Logical ERD contains 14 entities derived directly from the Functional Requirements in D2 SRS. All entities satisfy Third Normal Form (3NF). Descriptions below carry the same hedging language as D2 for any item that depends on confirmation by an external authority (Decree 288/2025/ND- CP article references, Plant Protection Dept report layout, VietGAP/GlobalGAP field set).

2.1 Entity Inventory

Entity (Table)	Key Attributes	Description	Linked FR
users	userId (PK), fullName, email, passwordHash, role, isActive, createdAt	Accounts for all 5 roles (RBAC). role = ENUM(ADMIN, HTX_MANAGER, OPERATOR, FARMER, AGRONOMIST). Admin is modelled via the role attribute only - no concrete Admin subclass.	FR18
cooperatives	cooperativeld (PK), name, province, address, maxDrones=50, managerId (FK→users)	HTX entity. maxDrones = 50 is a configurable reference workload / validation baseline (D1 v1.6 1.2), not a hard- coded system limit.	Scope (D1 1.2)
fields	fieldId (PK), farmerId (FK→users), cooperativeld (FK), fieldName, cropType, areaHa, gpsPolygon, createdAt	GIS polygon as GeoJSON; areaHa auto- computed via PostGIS ST_Area() on INSERT. UC09 BR: closed- ring polygon, 0.1–100 ha.	FR01, FR02
spray_formulas	formulaId (PK), fieldId (FK→fields), pesticideId (FK→pesticides), dosageLPerHa, sprayCountPerSeason, configuredBy (FK→users)	1:1 with fields. Agronomist configures the formula; per D2 v2.4 FR02 the final formula and dosage require confirmation by an authorized Agronomist or HTX Manager before mission dispatch (Agronomist.confirmDosage()).	FR02
drones	droneld (PK), cooperativeld (FK), serialNumber, model, tankCapacityL, batteryPercent, status (ENUM), gpsLat, gpsLng, updatedAt	Authoritative master record for FR21 (registered by HTX Manager/Admin via HTXManager.registerDrone()/updateDroneStatus()). Within MVP scope, gpsLat/gpsLng and batteryPercent come from the operator's mobile- app GPS (a mission- level proxy for drone position) or structured simulated telemetry, per D2 v2.4 2.5 - not native drone sensor telemetry (MAVLink is post-MVP).	FR06, FR10, FR11, FR21
operators	operatorId (PK), userId (FK→users), certificateNumber, certificateExpiry, operationalStatus	Pilot sub- profile. certificateExpiry validated on every assignment (UC02); expired- certificate pilots are hidden from suggestions (UC02 Alt 4a).	FR07
spray_orders	orderId (PK), farmerId (FK→users), fieldId (FK→fields), droneld (FK→drones, nullable), operatorId (FK→operators, nullable), scheduledDate, actualStartTime, status (ENUM), statusReason, pestAmountL, cancelReason, createdAt	Central transactional table. droneld/operatorId are NULL while status=PENDING (order created in UC01, assigned later in UC02). statusReason ∈ {"Weather", "Needs Review", "Lacking safety confirmation", "Confirmation timeout"} per the UC06 escalation logic.	FR03–FR09
flight_logs	logId (PK), orderId (FK→spray_orders, 1:1), droneld (FK→drones), operatorId	Compliance- support record for Decree 288/2025/ND- CP. gpsTrail = JSON array of {lat,lng,ts} every ~3s. manualOverride = TRUE marks a check- in approved by the HTX Manager	FR09, FR13

Entity (Table)	Key Attributes	Description	Linked FR
	(FK→operators), checkInTime, checkOutTime, checkInLat, checkInLng, pesticideUsedL, areaSprayed, gpsTrail (JSONB), manualOverride, syncedAt	under UC03 Alt 3a (GPS outside 100m) - per D2 v2.4 this requires HTX Manager approval, not pilot self- confirmation; the approval is additionally written to audit_logs. Checkout confirmation (UC07) is via the pilot's authenticated session (FR18 RBAC), not a biometric/PIN step.	
pesticides	pesticideId (PK), name, activeIngredient, isApprovedBVT, unit	BVT- approved master registry. isApprovedBVT is checked at booking (UC01 Alt 4a) and at order creation (FR03).	FR16
pesticide_stock	stockId (PK), cooperativeId (FK→cooperatives), pesticideId (FK→pesticides), stockL, lowStockThresholdL, updatedAt	Per- cooperative inventory, split from the pesticides master registry so stock can vary across cooperatives sharing the same approved list. Deducted after a COMPLETED spraying mission.	FR16
geofences	geofenceId (PK), fieldId (FK→fields, 1:1), boundaryPolygon (GeoJSON), bufferMeters=50, hazardZones (JSONB), isActive	Safety boundary. bufferMeters=50 is the default buffer around registered high- voltage power lines, informed by the Long An 110kV incident (Tuoi Tre, 2024; D1 v1.6 [1]). Per D2 v2.4 FR12, the system issues information- level alerts only when a drone approaches or crosses this boundary - it does not directly control the drone, firmware, or autopilot; the pilot remains responsible for flight safety.	FR12
notifications	notificationId (PK), recipientId (FK→users), relatedOrderId (FK→spray_orders, nullable), channel, eventType, content, status, sentAt	Outbound message log. channel ∈ {SMS, ZALO, PUSH}. Triggers: booking received/confirmed, spraying started/completed/suspended, alerts.	FR17
audit_logs	auditId (PK), eventType, actorId (FK→users), actorRole, targetEntityId, targetEntityType, description, ipAddress, occurredAt	Append- only audit trail (never updated or deleted). Covers weather overrides (UC01/UC06), report exports (UC08), data- sharing access approvals (UC10), and GPS check- in overrides (UC03 Alt 3a). Supports NFR05's "all security- relevant events written to an immutable audit log" outcome.	NFR05, FR13
reports	reportId (PK), generatedBy (FK→users), cooperativeId (FK→cooperatives), reportType, periodStart, periodEnd, fileFormat, fileUrl, exportedAt	Compliance- support export record. reportType ∈ {DECREE_288, VIETGAP, GLOBALGAP}. Per D2 v2.4, the report layout is aligned with Plant Protection Dept reporting fields, but the exact official format is subject to authority confirmation (FR14); the VietGAP/GlobalGAP field set is a project- defined minimum, pending stakeholder validation with contracted export companies (FR15).	FR14, FR15

2.2 Relationships & Cardinality

From Entity	Cardinality	To Entity	FK Column	Rationale
users	1: 0..*	spray_orders (as Farmer)	farmerId	One farmer places many spray orders
users	1: 0..1	operators	userId	A user may optionally hold an Operator sub- profile
users	1: 0..*	reports	generatedBy	A user (HTX Manager) generates many reports
cooperatives	1: 1..*	drones	cooperativeld	One cooperative manages 1..* drones (≤ 50 , configurable)
cooperatives	1: 0..*	pesticide_stock	cooperativeld	Per- cooperative pesticide inventory
fields	1: 1	spray_formulas	fieldId	Each field has exactly one spray formula
fields	1: 0..*	spray_orders	fieldId	One field can have many orders across seasons
fields	1: 1	geofences	fieldId	Each field has exactly one safety geofence
drones	1: 0..*	spray_orders	droneId (nullable)	A drone may be assigned to many orders over time
spray_orders	1: 0..1	flight_logs	orderId	A completed/suspended/cancelled order generates at most one log
flight_logs	0.* : 1..*	reports	junction: report_flight_logs	A report aggregates ≥ 1 flight logs; a log may appear in 0..* reports
pesticides	1: 0..*	spray_formulas	pesticideId	One pesticide may be used in many formulas
pesticides	1: 0..*	pesticide_stock	pesticideId	One pesticide is stocked across many cooperatives
spray_orders	1: 0..*	notifications	relatedOrderId (nullable)	One order may trigger many notifications
users	1: 0..*	audit_logs	actorId	One user is the actor in many audit events

2.3 Normalization Proof (3NF)

Normal Form	Rule	How Satisfied in AgriUAV	Status
1NF	All attributes are atomic (no repeating groups).	gpsTrail, hazardZones stored as JSONB (single atomic column); gpsPolygon as a single GeoJSON string.	Satisfied
2NF	No partial dependency on a composite key.	All tables use single- column surrogate UUID v4 primary keys. No composite PKs exist.	Trivially satisfied
3NF	No transitive dependency: every non- key attribute depends only on the PK.	spray_orders does not store cooperativeld (derived via drones/fields), pilot name (via operatorId→users), or field area (via fieldId→fields).	Satisfied

2.4 FR21 Modeling Decision - “No Standalone Use Case”

FR21 (“HTX Manager and Admin register and manage drone master records”) is implemented entirely through the existing drones entity and the HTXManager.registerDrone()/updateDroneStatus() methods on the Class Diagram. Per the D2 traceability note (5.1) and the audit reasoning in Entry #20, FR21 is treated as a prerequisite data- management function rather than a standalone use case: it is read by UC02 (drone status/availability check for assignment) and UC08 (drone identifier listing in the compliance report). No UC11 is introduced in the Use Case Diagram - the use- case set stays at 10 UCs, matching D2 and UseCaseDiagram

3. Physical Data Model

Technology stack: PostgreSQL 16 + PostGIS 3.4 (spatial queries) + TimescaleDB 2.x (time- series telemetry). All UUIDs are v4. HYPERTABLE rows indicate TimescaleDB time- series partitioning. Retention note: flight_logs, drone_positions, and audit_logs are retained for at least 5 years as a working assumption pending confirmation of the specific Decree 288/2025/ND- CP article reference (per D2 | 2.4) - not asserted here as a confirmed legal minimum.

Table: users

Column	Data Type	Null	Key	Default	Description
userId	VARCHAR(36)	NO	PK	UUID v4	Primary key
fullName	VARCHAR(255)	NO			Full display name
email	VARCHAR(255)	NO	UNIQUE		Login email
passwordHash	VARCHAR(255)	NO			bcrypt hash (cost 12)
role	ENUM(5)	NO			ADMIN HTX_MANAGER OPERATOR FARMER AGRONOMIST
isActive	BOOLEAN	NO		TRUE	Soft delete flag
createdAt	TIMESTAMP	NO		NOW()	Account creation UTC

Table: cooperatives

Column	Data Type	Null	Key	Default	Description
cooperativeld	VARCHAR(36)	NO	PK	UUID v4	
name	VARCHAR(255)	NO			Official cooperative name
province	VARCHAR(100)	NO			Province in the Mekong Delta
address	TEXT	YES			
maxDrones	INTEGER	NO		50	Configurable reference workload, not a hard limit (D1 v1.6 1.2)
managerId	VARCHAR(36)	NO	FK→users		HTX Manager account

Table: fields

Column	Data Type	Null	Key	Default	Description
fieldId	VARCHAR(36)	NO	PK	UUID v4	
farmerId	VARCHAR(36)	NO	FK→users		Owner farmer
cooperativeld	VARCHAR(36)	NO	FK→cooperatives		Managing cooperative
fieldName	VARCHAR(255)	NO			
cropType	VARCHAR(100)	NO			e.g. rice, corn
areaHa	DECIMAL(8,4)	NO			Auto- computed from gpsPolygon (PostGIS ST_Area)
gpsPolygon	TEXT	NO			GeoJSON polygon - GIST index
createdAt	TIMESTAMP	NO		NOW()	

INDEX (gpsPolygon) - PostGIS GIST index for spatial queries.

Table: drones

Column	Data Type	Null	Key	Default	Description
droneId	VARCHAR(36)	NO	PK	UUID v4	
cooperativeld	VARCHAR(36)	NO	FK→cooperatives		Owner cooperative
serialNumber	VARCHAR(100)	NO	UNIQUE		Decree 288/2025 drone identifier (FR21)
model	VARCHAR(100)	NO			e.g. DJI Agras T40
tankCapacityL	DECIMAL(5,2)	NO			
batteryPercent	SMALLINT	NO		100	Mobile- GPS proxy or simulated telemetry (MVP)
status	ENUM(4)	NO		IDLE	IDLE FLYING MAINTENANCE OFFLINE
gpsLat	DECIMAL(9,6)	YES			Mobile- GPS proxy or simulated telemetry (MVP)
gpsLng	DECIMAL(9,6)	YES			Mobile- GPS proxy or simulated telemetry (MVP)
updatedAt	TIMESTAMP	NO		NOW()	Auto- updated on WebSocket push

INDEX (status) - B- Tree index for availability queries. FR21:

HTXManager.registerDrone()/updateDroneStatus() write to this table.

Table: operators

Column	Data Type	Null	Key	Default	Description
operatorId	VARCHAR(36)	NO	PK	UUID v4	
userId	VARCHAR(36)	NO	FK→users, UNIQUE		
certificateNumber	VARCHAR(50)	NO	UNIQUE		Decree 288/2025 pilot certificate
certificateExpiry	DATE	NO			Validated on every assignment (UC02)
operationalStatus	VARCHAR(50)	NO		'ACTIVE'	ACTIVE ON_LEAVE SUSPENDED

INDEX (certificateExpiry) - filters expired- certificate pilots from assignment suggestions (UC02 Alt 4a).

Table: spray_formulas

Column	Data Type	Null	Key	Default	Description
formulaId	VARCHAR(36)	NO	PK	UUID v4	
fieldId	VARCHAR(36)	NO	FK→fields, UNIQUE		1:1 with fields
pesticideId	VARCHAR(36)	NO	FK→pesticides		
dosageLPerHa	DECIMAL(6,2)	NO			Dosage per hectare (FR02)
sprayCountPerSeason	SMALLINT	NO			
configuredBy	VARCHAR(36)	NO	FK→users		Agronomist who configured the formula

Table: spray_orders

Column	Data Type	Null	Key	Default	Description
orderId	VARCHAR(36)	NO	PK	UUID v4	
farmerId	VARCHAR(36)	NO	FK→users		
fieldId	VARCHAR(36)	NO	FK→fields		
droneId	VARCHAR(36)	YES	FK→drones	NULL	NULL until assigned (UC02)
operatorId	VARCHAR(36)	YES	FK→operators	NULL	NULL until assigned
scheduledDate	TIMESTAMP	NO			Requested spray datetime
actualStartTime	TIMESTAMP	YES		NULL	Set on check- in
status	ENUM(6)	NO		PENDING	PENDING→ASSIGNED→IN_PR GRESS→COMPLETED SUSP ENDED CANCELLED
statusReason	TEXT	YES		NULL	“Weather” “Needs Review” “Lacking safety confirmation” “Confirmation timeout”
pestAmountL	DECIMAL(6,2)	YES		NULL	Calculated on assignment
cancelReason	TEXT	YES		NULL	
createdAt	TIMESTAMP	NO		NOW()	

INDEX (fieldId, scheduledDate) - conflict checks (FR04). INDEX (status) - active- order filtering. “No-assign” path keeps status = PENDING (UC02 Alt 3a), not CANCELLED.

Table: flight_logs

Column	Data Type	Null	Key	Default	Description
logId	VARCHAR(36)	NO	PK	UUID v4	
orderId	VARCHAR(36)	NO	FK→spray_orders, UNIQUE		1:1 with spray_orders
droneId	VARCHAR(36)	NO	FK→drones		
operatorId	VARCHAR(36)	NO	FK→operators		
checkInTime	TIMESTAMP	NO			
checkOutTime	TIMESTAMP	YES		NULL	
checkInLat	DECIMAL(9,6)	NO			
checkInLng	DECIMAL(9,6)	NO			
pesticideUsedL	DECIMAL(6,2)	YES		NULL	
areaSprayed	DECIMAL(8,4)	YES		NULL	
gpsTrail	JSONB	YES		NULL	{lat,lng,ts} approx. every 3s
manualOverride	BOOLEAN	NO		FALSE	TRUE only after HTX Manager approval of a GPS- out- of- 100m check- in (UC03 Alt 3a)
syncedAt	TIMESTAMP	YES		NULL	Offline sync timestamp (NFR06, ≤30s)

HYPERTABLE (checkInTime), partitioned monthly. INDEX (droneId, checkInTime). Checkout confirmation (UC07) is via the pilot's authenticated session (FR18), not a biometric/PIN field.

Table: drone_positions (derived - supports FR10/NFR01, not one of the 14 logical entities)

Column	Data Type	Null	Key	Default	Description
posId	BIGSERIAL	NO	PK	AUTO	Auto- increment for high- volume inserts
droneId	VARCHAR(36)	NO	FK→drones		
orderId	VARCHAR(36)	YES	FK→spray_orders	NULL	NULL if idle telemetry
lat	DECIMAL(9,6)	NO			
lng	DECIMAL(9,6)	NO			
recordedAt	TIMESTAMP	NO		NOW()	

*HYPERTABLE (recordedAt), partitioned weekly; retention per the working 5- year assumption above.
INDEX (droneId, recordedAt DESC) - latest- position queries for the live map.*

Table: pesticides

Column	Data Type	Null	Key	Default	Description
pesticideId	VARCHAR(36)	NO	PK	UUID v4	
name	VARCHAR(255)	NO			
activeIngredient	VARCHAR(255)	NO			
isApprovedBVT V	BOOLEAN	NO		FALSE	
unit	VARCHAR(20)	NO		'L'	

INDEX (isApprovedBVT) - filters the booking dropdown to approved pesticides.

Table: pesticide_stock

Column	Data Type	Null	Key	Default	Description
stockId	VARCHAR(36)	NO	PK	UUID v4	
cooperativeld	VARCHAR(36)	NO	FK→cooperatives		
pesticideId	VARCHAR(36)	NO	FK→pesticides		
stockL	DECIMAL(8,2)	NO		0	
lowStockThreshold L	DECIMAL(8,2)	NO			
updatedAt	TIMESTAMP	NO		NOW()	

UNIQUE (cooperativeld, pesticideId) - one stock record per pesticide per cooperative.

Table: geofences

Column	Data Type	Null	Key	Default	Description
geofenceId	VARCHAR(36)	NO	PK	UUID v4	
fieldId	VARCHAR(36)	NO	FK→fields, UNIQUE		
boundaryPolygon	TEXT	NO			GeoJSON - GIST index
bufferMeters	INTEGER	NO		50	Default buffer around registered 110kV lines (FR12)
hazardZones	JSONB	YES		NULL	{type:'POWERLINE' 'RESIDENTIAL', coords}
isActive	BOOLEAN	NO		FALSE	Set TRUE on UC03 check- in

Alert behaviour: information- level only - the system never directly controls the drone, firmware, or autopilot.

Table: notifications

Column	Data Type	Null	Key	Default	Description
notificationId	VARCHAR(36)	NO	PK	UUID v4	
recipientId	VARCHAR(36)	NO	FK→users		
relatedOrderId	VARCHAR(36)	YES	FK→spray_orders	NULL	
channel	VARCHAR(20)	NO			SMS ZALO PUSH
eventType	VARCHAR(50)	NO			
content	TEXT	NO			
status	VARCHAR(20)	NO		'PENDING'	PENDING SENT FAILED
sentAt	TIMESTAMP	YES		NULL	

INDEX (recipientId, sentAt DESC) - delivery history.

Table: audit_logs

Column	Data Type	Null	Key	Default	Description
auditId	VARCHAR(36)	NO	PK	UUID v4	
eventType	ENUM(7)	NO			ORDER_CREATED ORDER_CANCELLED WEATHER_OVERRIDE REPORT_EXPORTED DATA_ACCESS CHECKIN CHECKOUT
actorId	VARCHAR(36)	NO	FK→users		
actorRole	ENUM(5)	NO			

Column	Data Type	Null	Key	Default	Description
targetEntityId	VARCHAR(36)	NO			
targetEntityType	VARCHAR(50)	NO			
description	TEXT	NO			
ipAddress	VARCHAR(45)	YES		NULL	
occurredAt	TIMESTAMP	NO		NOW()	

Append- only (DB- level trigger blocks UPDATE/DELETE). HYPERTABLE (occurredAt). INDEX (actorId, occurredAt DESC). GPS check- in overrides (UC03 Alt 3a) are logged here with eventType = CHECKIN.

Table: reports

Column	Data Type	Null	Key	Default	Description
reportId	VARCHAR(36)	NO	PK	UUID v4	
generatedBy	VARCHAR(36)	NO	FK→users		
cooperativeld	VARCHAR(36)	NO	FK→cooperatives		
reportType	ENUM(3)	NO			DECREE_288 VIETGAP GLOBALGAP
periodStart	DATE	NO			
periodEnd	DATE	NO			
fileFormat	VARCHAR(10)	NO			PDF XLSX
fileUrl	TEXT	NO			
exportedAt	TIMESTAMP	NO		NOW()	

DECREE_288 layout aligned to Plant Protection Dept fields, exact official format pending authority confirmation (FR14). VietGAP/GlobalGAP field set is project- defined and pending export- company validation (FR15).

4. Data Dictionary - Key Fields Reference

Summarises the most business- critical fields across all tables - those directly implementing a Functional Requirement, a Decree 288/2025/ND- CP compliance- support obligation, or a consistency fix.

Table	Field	Data Type	Constraint	Business Rule / Source
users	role	ENUM	NOT NULL	Controls RBAC (FR18); 5 values incl. AGRONOMIST. ADMIN has no concrete subclass - full access via role check only.
fields	gpsPolygon	TEXT	NOT NULL	Closed- ring GeoJSON; area must be 0.1–100 ha (UC09 BR).
fields	areaHa	DECIMAL (8,4)	NOT NULL	Computed via PostGIS ST_Area() on INSERT/UPDATE; never entered manually.
drones	status	ENUM	NOT NULL	IDLE→FLYING (assignment) →IDLE (check- out) →MAINTENANCE (manual); OFFLINE after >15s no signal (UC04 Alt 2a).
drones	batteryPercent / gpsLat / gpsLng	SMALLINT / DECIMAL	NOT NULL / NULLABLE	Within MVP scope, sourced from the operator's mobile- GPS proxy or structured simulated telemetry (D2 v2.4 2.5); native MAVLink sensor telemetry is post- MVP.
spray_orders	status	ENUM	NOT NULL	PENDING→ASSIGNED→IN_PROGRESS→COMPLETED SUSPENDED CANCELLED; each transition timestamped.
spray_orders	statusReason	TEXT	NULLABLE	“Weather” / “Needs Review” / “Lacking safety confirmation” / “Confirmation timeout” - set by the UC06 weather- escalation logic.
spray_orders	droneId / operatorId	VARCHAR (FK)	NULLABLE	NULL while PENDING - orders are created (UC01) before assignment (UC02); the “no- assign” path keeps status = PENDING, not CANCELLED.
operators	certificateExpiry	DATE	NOT NULL	Validated on every assignment (UC02); expired pilots are hidden from suggestions (UC02 Alt 4a).
flight_logs	gpsTrail	JSONB	NULLABLE	{lat,lng,ts} approx. every 3s; compliance- support evidence, not a legally confirmed minimum interval.
flight_logs	manualOverride	BOOLEAN	NOT NULL DEFAULT FALSE	FIX: TRUE only after HTX Manager approval of a GPS- out-of- 100m check- in (UC03 Alt 3a). The Pilot can no longer self-confirm an override; approval is also written to audit_logs (eventType = CHECKIN).
flight_logs	checkout confirmation (process rule, not a column)	—	—	FIX: Confirmed via the Pilot's authenticated session (FR18 RBAC) per UC07 step 4 - not a biometric/PIN field. An earlier draft's fingerprint/PIN requirement was removed in D2 v2.4 as a hidden requirement.
pesticides	isApprovedBVTV	BOOLEAN	NOT NULL	FALSE blocks order confirmation (UC01 Alt 4a) and hides the pesticide from the booking dropdown.
pesticide_stock	stockL / lowStockThresholdL	DECIMAL	NOT NULL	Deducted after a COMPLETED mission; low- stock alert when stockL < lowStockThresholdL (FR16).
geofences	bufferMeters	INTEGER DEFAULT 50	NOT NULL	Default 50m buffer around registered 110kV lines, informed by the Long An incident (Tuoi Tre, 2024); configurable per field.

Table	Field	Data Type	Constraint	Business Rule / Source
geofences	alert behaviour (process rule, not a column)	—	—	FIX: information- level alert only when a drone approaches/crosses the boundary; the system never directly controls the drone, firmware, or autopilot (FR12). "VIOLATION" wording removed throughout this document.
notifications	channel	VARCHAR	NOT NULL	SMS ZALO PUSH; decoupled from spray_orders for multi-channel delivery tracking (FR17).
audit_logs	eventType / occurredAt	ENUM / TIMESTAMP	NOT NULL	Append- only; never updated or deleted; supports NFR05's immutable- audit- log outcome.
reports	reportType	ENUM	NOT NULL	DECREE_288 (FR14, layout aligned to PPD fields, exact official format pending authority confirmation) VIETGAP/GLOBALGAP (FR15, project- defined minimum field set pending export- company validation).
drone_positions	recordedAt	TIMESTAMP	NOT NULL	TimescaleDB chunk key; retention follows the working 5- year assumption (I3), pending confirmation of the specific Decree 288 article reference.

5. Traceability Matrix - FR ↔ Model Elements

Maps all 21 Functional Requirements from D2 SRS (FR01–FR21, including the new FR21 drone registry requirement) to the model elements in D3: Class/Entity, DB Table, Use Case, and Diagram. The Diagram column states exactly which diagrams cover each FR, including flags where the diagram itself still needs a correction before it can be considered fully synchronized.

5.1 Full Matrix

FR ID	Short Description	Priority	Class/Entity	DB Table	UC	Diagram
FR01	Create/edit field maps	Must	Field	fields	UC09	Class Diagram, AD1
FR02	Calculate per- zone dosage + Agronomist/HTX confirmation	Must	Field, SprayFormula, Agronomist	fields, spray_formulas	UC09	Class Diagram, AD1
FR03	Book spraying ≤ 5 steps	Must	SprayOrder	spray_orders	UC01	Class Diagram, Sequence UC01, AD1
FR04	Schedule- conflict alerts	Must	SprayOrder	spray_orders (idx fieldId+date)	UC01	Class Diagram, Sequence UC01 Alt3a, AD1
FR05	Calendar view (HTX Manager)	Should	HTXManager, SprayOrder	spray_orders	UC01, UC02 (indirect)	Class Diagram (viewCalendar())
FR06	Rule- based drone–pilot assignment recommendation	Must	Drone, SprayOrder	drones, spray_orders	UC02	Class Diagram, AD1
FR07	Pilot records (certificate)	Must	Operator	operators	UC02, UC08	Class Diagram, AD1
FR08	Assignment notification to Operator	Must	SprayOrder, Notification	spray_orders, notifications	UC02	Class Diagram, AD1
FR09	GPS check- in / check- out	Must	FlightLog, Operator	flight_logs	UC03, UC07	Class Diagram, Sequence UC03 △, AD2 (UC07)
FR10	Near real- time map ≤ 3s (telemetry proxy)	Must	Drone	drone_positions	UC03, UC04, UC05	Class Diagram, Sequence UC04, AD2
FR11	Battery/pesticide/geofence/ weather info- level alerts	Must	Drone, Geofence	drones, geofences	UC04, UC05, UC06	Class Diagram, Sequence UC04 Alt5a, AD2
FR12	Geofencing - 50m buffer, information- level only	Must	Geofence	geofences	UC03, UC04	Class Diagram, Sequence UC03 Alt3a △, Sequence UC04 Alt5a, AD2
FR13	Complete flight logs (compliance- support)	Must	FlightLog, AuditLog	flight_logs, drone_positions , audit_logs	UC02, UC04, UC07, UC08	Class Diagram, Sequence UC03/UC04, AD1, AD2
FR14	Export Plant Protection Dept report (format pending confirmation)	Must	Report	reports	UC08, UC10	Class Diagram, ERD 3

FR ID	Short Description	Priority	Class/Entity	DB Table	UC	Diagram
FR15	Export VietGAP/GlobalGAP (project- defined minimum field set)	Should	Report	reports	UC08, UC10	Class Diagram, ERD 3
FR16	Pesticide inventory management	Should	Pesticide, PesticideStock	pesticides, pesticide_stock	UC01, UC07	Class Diagram, AD2 (stock deduction)
FR17	SMS/Zalo notifications	Should	Notification	notifications	UC01, UC06, UC07	Class Diagram, Sequence UC01 Alt, AD1, AD2
FR18	RBAC - 5 roles	Must	User (abstract)	users (role ENUM)	All UCs (cross-cutting)	Class Diagram, Use Case Diagram (Admin generalization)
FR19	Dark mode UX	Could	—	—	Cross-cutting UX	No dedicated diagram
FR20	Multi- language VI/EN	Could	—	—	Cross-cutting i18n	No dedicated diagram
FR21	Drone master record registry	Must	Drone, HTXManager	drones	UC02, UC08 (prerequisite read - no standalone UC, 2.4)	Class Diagram

5.2 Coverage Summary

Sequence and Activity Diagram percentages are computed directly from the Linked- FR list declared in each diagram's own header (AD1_BookingFlow, AD2_MonitoringAlertFlow) or the Sequence Diagram sections carried forward from D3 old version not estimated. They will need to be re-verified once Sequence Diagram UC03 is redrawn (1.8).

Coverage Dimension	Detail	Count	Coverage %	Note
Total FRs	21 FRs (15 Must, 4 Should, 2 Could)	21/21	100%	All FRs documented in D2 SRS v2.4
Traced to Class Diagram	19/21 FRs (FR19, FR20 are cross- cutting UX/i18n)	19/21	≈90%	FR19, FR20 explicitly excluded - no class represents UI theme or i18n
Traced to DB Table	19/21 FRs	19/21	≈90%	Same exclusions as Class Diagram
Traced to Sequence Diagram	FR03,04,09,10,11,12,13,16,17 (per Sequence UC01/UC03/UC04 headers)	9/21	≈43%	UC09 (FR01,02) and UC08/UC10 (FR14,15) have no dedicated Sequence Diagram
Traced to Activity Diagram	FR01,02,03,04,06,07,08,09,10,11,12,13,16,17,18 (per AD1_v3/AD2_v3 headers)	15/21	≈71%	FR05,14,15,19,20,21 are not in either AD1 or AD2's declared FR list
Traced to ≥1 UC	19/21 FRs (Direct / Indirect / Cross- cutting / Prerequisite per D2 v2.4 5.1 Coverage Type)	19/21	≈90%	FR19, FR20 are not tied to any UC

6. AI Artifact Audit Log - Entries #14 - 20

Entries #14 - 19 are carried forward unchanged from D3 (the historical record of AI interactions is not rewritten retroactively). Legend: Accepted | Rejected | Refined | Added | Hallucination/Bias detected | Open / not yet resolved.

Entry #14 - W5

Field	Content
DTC Component	Abstraction
AI Output (tóm tắt)	AI đề xuất 6 sub- processes cho DFD Level 1: (1) Field Mgmt, (2) Scheduling, (3) Fleet Dispatch, (4) Monitoring, (5) Notification, (6) User Management. 8 external entities.
Decision & Refinement	Chấp nhận 4 sub- processes cốt lõi: Field Mgmt, Scheduling, Monitoring, Compliance. Loại Notification và User Mgmt - đây là cross- cutting services, không phải standalone process ở L1 theo Gane- Sarson convention. AI không phân biệt được cross- cutting vs standalone process - Human Delta quan trọng.
Verification	Cross- check DFD balanced rule - tất cả data flow vào/ra Level 0 phải xuất hiện đúng ở Level 1.
AI Tool / Prompt	Claude - “Đề xuất external entities, data flows và sub- processes cho DFD Level 1 của AgriUAV Platform.”

Entry #15 - W6

Field	Content
DTC Component	Abstraction
AI Output (tóm tắt)	AI sinh Drone class với 25+ attributes (firmware_version, sensor_list, obstacle_avoidance_level...) và 18 methods bao gồm cả low- level hardware ops.
Decision & Refinement	Chấp nhận 9 core attributes cho Drone: droneId, serialNumber, model, tankCapacityL, batteryPercent, gpsLat, gpsLng, status, cooperativeId. Loại 16 attributes còn lại - thuộc phạm vi drone manufacturer SDK, không thuộc IS database. Gộp 5 methods phức tạp thành 3 core: isAvailable(), updatePosition(), sendAlert().
Verification	Kiểm tra với FR06 (dispatch cần status), FR10 (monitoring cần gpsLat/Lng), FR11 (alert cần batteryPercent) - 9 attributes đủ để fulfill tất cả FR liên quan.
AI Tool / Prompt	Claude - “Vẽ Class Diagram cho AgriUAV Platform với đầy đủ attributes và methods.”

Entry #16 - W6

Field	Content
DTC Component	Process / Algorithm
AI Output (tóm tắt)	AI sinh sequence cho UC03 với 5 messages (happy path only). Không có alternative flows, không có offline scenario.
Decision & Refinement	Chấp nhận happy path cơ bản. Bổ sung ALT 3a: GPS outside 100m radius → manual override với reason (FR09 business rule). Bổ sung ALT 4a: Offline mode (no 4G) → local cache → auto- sync ≤30s khi reconnect (NFR06). v1.4: chuẩn hóa thuật ngữ cache offline thành IndexedDB.
Verification	Cross- check với D2 SRS NFR06 (offline ≤30s sync, IndexedDB) và UC03 Alt flows.
AI Tool / Prompt	Claude - “Tạo Sequence Diagram cho UC03 - On- site GPS Check- in.”

Entry #17 - W6

Field	Content
DTC Component	Process / Algorithm
AI Output (tóm tắt)	AI đề xuất activity diagram tuyến tính (sequential) cho AD2: check geofence → check battery → check weather → update map. Không có fork/join, escalation timers, repeat loop, swim lanes.
Decision & Refinement	PHẢN BIỆN: Sequential diagram sai về mặt kỹ thuật - geofence alert có thể bị delay khi đang wait weather API response (drone bay 5–7 m/s, 1s = 5–7m drift). Thiết kế lại với fork/join 3 checks đồng thời, escalation timers (5 phút UC05, 10 phút UC06, force- suspend 30 phút), repeat loop, swim lanes.
Verification	Cross- check với UC05/UC06 escalation trong D2 SRS. Tham chiếu DJI Agras T40: tốc độ bay tối đa 7 m/s.
AI Tool / Prompt	Claude - “Thiết kế Activity Diagram cho quy trình monitoring drone đang bay, 3 loại alert, escalation timers, parallel processing.”

Entry #18 - W8

Field	Content
DTC Component	Pattern Recognition
AI Output (tóm tắt)	AI kết luận Microservices ‘tốt nhất cho scalability’ với reasoning generic (Netflix/Uber style). Event- Driven xếp hạng 2.
Decision & Refinement	AI bias toward ‘trending architecture’ - Microservices overhead không phù hợp scope HTX ĐBSCL (1 cooperative = 50 drones = 5 modules; HTX không có DevOps team; inter- service latency ~50–100ms không phù hợp NFR01 ≤3s; team 5 SV/10 tuần). Final choice: Layered Architecture.
Verification	NFR01 (≤3s), NFR04 (50- drone reference workload), NFR07 (maintainability) - Layered thỏa mãn tất cả.
AI Tool / Prompt	Claude - “So sánh Layered vs Microservices vs Event- Driven cho AgriUAV HTX ĐBSCL ~50 drones.”

Entry #19 - W9

Field	Content
DTC Component	Process / Algorithm
AI Output (tóm tắt)	AI đề xuất 6 STRIDE threats generic (Spoofing, Tampering, Repudiation, Info Disclosure, DoS, EoP) với mitigation generic (JWT, HTTPS, logging, rate limiting).
Decision & Refinement	STRIDE framework. Tinh chỉnh từng threat: Spoofing → ‘Fake GPS check- in’ (100m radius + manualOverride flag); Tampering → ‘WebSocket MITM inject false telemetry’ (TLS 1.3 + message signing); Repudiation → append- only FlightLog + DB audit trigger; Information Disclosure → gpsTrail AES- 256 + RBAC field- level; EoP → JWT role tampering (server- side validation).
Verification	OWASP Top 10, NFR05 (TLS 1.3, AES- 256, JWT), Decree 288/2025/NĐ- CP append- only audit requirement (working assumption).
AI Tool / Prompt	Claude - “STRIDE threat model AgriUAV: airspace data, PII nông dân, telemetry, compliance logs.”

Entry #20 - W6

Field	Content
DTC Component	Pattern Recognition / Consistency Audit
AI Output (tóm tắt)	Claude rà soát D3 v1.4 đối chiếu với D1 v1.6 và D2 v2.4 (đã có các bản sửa UC10 actor, UC03 Alt3a, UC07 bỏ biometric, telemetry semantics, FR21 mới, 5 roles). Phát hiện 5 nhóm vấn đề đồng bộ: (1) Use Case Diagram vẫn chưa tồn tại như một artifact riêng; (2) DFD Level 1 thiếu rationale khi loại Operations Dept, đã từng mất 5 data flow (n1–n5); (3) Class Diagram chưa có method cho FR21 và confirmDosage() cho FR02; (4) ngôn ngữ “GEOFENCE VIOLATION” mâu thuẫn với “information- level alert” của FR12; (5) Sequence Diagram UC03 Alt 3a vẫn cho phép Pilot tự xác nhận override GPS - mâu thuẫn với D2 v2.4.
Decision & Refinement	Tạo Use Case Diagram mới (UseCaseDiagram_v1.5.puml) - đóng gap (1). Cập nhật DFD (level0_v1.5, level1_v1.5): loại Operations Dept có rationale; khôi phục n1–n5; Export Company → “post- MVP” - đóng gap (2). Cập nhật Class Diagram (v1.5): thêm registerDrone()/updateDroneStatus() (FR21), confirmDosage() (FR02), requestCheckInOverride() - đóng gap (3) và một phần gap (5). Chuẩn hóa “VIOLATION” → “information- level alert” trong core content - đóng một phần gap (4). Sequence UC03 CHƯA được vẽ lại (theo yêu cầu Minh - UML để trống, tự paste sau) - ghi thành outstanding item, KHÔNG tự ý coi là đã đóng. Xác nhận “không có UC11” cho FR21 (giữ nguyên từ D2 v2.4 5.1). Viết lại Traceability Matrix cho 21 FR, % coverage tính trực tiếp từ Linked- FR header của từng diagram.
Verification	Đối chiếu trực tiếp với D1 v1.6, D2 v2.4 (FR01–FR21, UC01–UC10, NFR04/NFR05 hedge), ClassDiagram_v1.5.puml, UseCaseDiagram_v1.5.puml, DFD_Agriuv_level0_v1.5.drawio, DFD_Agriuv_level1_v1.5.drawio (đã có sẵn trước khi audit này thực hiện). Sequence/Activity FR nội dung tham chiếu theo header đã công bố trong AD1_v3/AD2_v3.
AI Tool / Prompt	Claude - “Audit D3 v1.4 against D1 v1.6 and D2 v2.4; identify structural/consistency gaps; produce updated D3 v1.5 core content while leaving UML/DFD diagram sections as placeholders.”

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- [7] Thanh Nien Newspaper. (June 2025). Province-Level Statistical Breakdown of Active Agricultural Drone Fleets in the Mekong Delta Region. <https://thanhvien.vn/bung-no-uav-nong-nghiep-o-mien-tay-185250617232615926.htm>
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